Preliminary Mechanical Design of a Simple Robotic Dog

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1. Introduction

Quadruped robots are widely used in robotics because they provide excellent stability and mobility compared to wheeled robots, particularly on uneven terrain. Their mechanical design requires careful planning of the body structure, joints, actuators, and balance to ensure efficient movement.

This project presents the mechanical design of a simple robotic dog developed using **Tinkercad**. The main objective is to create a basic four-legged robotic structure that demonstrates the fundamental concepts of quadruped robot design while keeping the model simple and suitable for beginners.

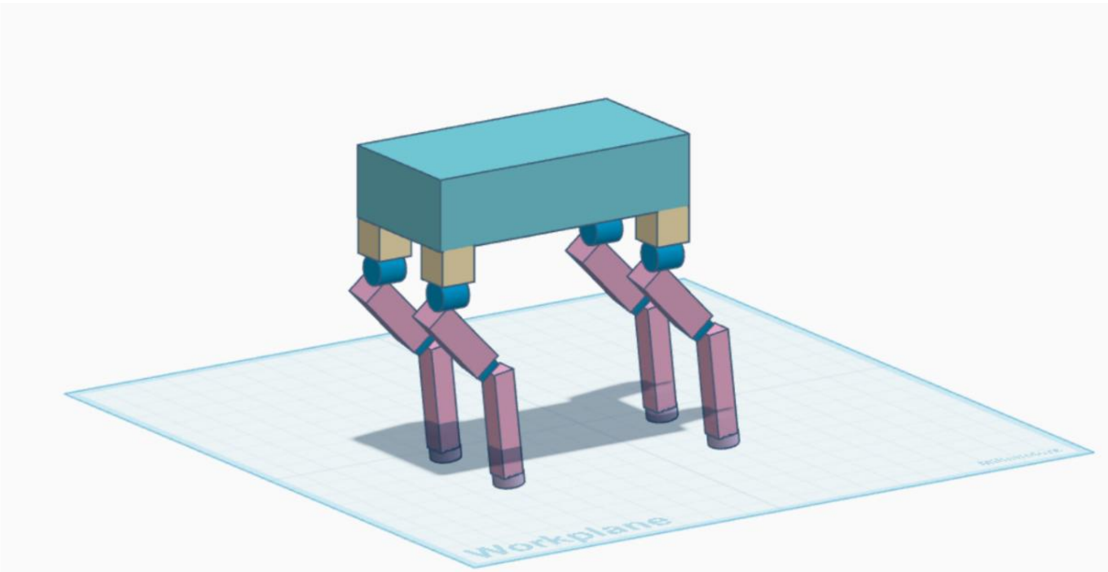
2. Project Objectives

The objectives of this project are:

- Design a simple robotic dog chassis.
- Design four identical legs.
- Define the joints and Degrees of Freedom (DOF).
- Select appropriate motors.
- Estimate the required joint torque.
- Analyze stability and center of gravity.
- Propose a walking method.
- Identify expected mechanical challenges.

3. Mechanical Design

The robot consists of a rectangular body supported by four identical legs. Each leg includes a hip joint, an upper leg, a knee joint, and a lower leg. The model was created using basic geometric shapes available in Tinkercad, making the design simple while demonstrating the essential mechanical components of a robotic dog.



4. Body and Chassis Design

The chassis was designed using a rectangular box with approximate dimensions of **80 × 40 × 20 mm**. Small mounting blocks were added at the four corners of the body to support the hip joints.

The rectangular chassis was selected because it:

- Provides a simple and lightweight structure.
- Offers sufficient space for future electronic components.
- Improves balance through symmetrical weight distribution.
- Is easy to manufacture and modify.

The symmetrical body layout helps maintain a stable center of gravity during movement.

5. Leg Design

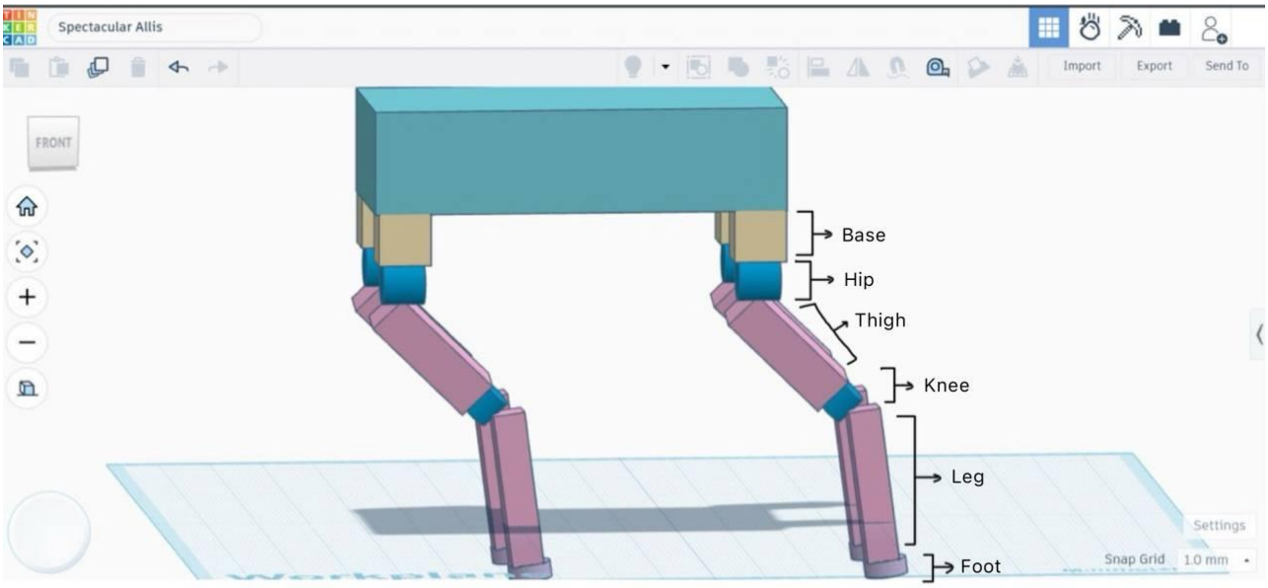
Each of the four legs has the same mechanical structure.

Each leg consists of:

- Hip Joint
- Upper Leg (Thigh)
- Knee Joint
- Lower Leg (Shin)

The hip and knee joints are represented by cylindrical components, while the upper and lower legs are represented by rectangular blocks.

Using identical legs simplifies the mechanical design and improves stability.



6. Joints and Degrees of Freedom

Each leg contains two rotational joints:

- Hip Joint (1 DOF)
- Knee Joint (1 DOF)

Therefore:

Component	Quantity
legs	4
Joints per Leg	2
Total Joints	8
Total Degrees of Freedom	DOF 8

Each joint provides one rotational degree of freedom, allowing the robot to perform basic walking movements.

7. Motor Selection

The selected actuator for this project is the **SG90 Servo Motor**.

The SG90 was selected because it:

- Is lightweight.
- Has a compact size.
- Is easy to control.
- Has low power consumption.
- Is widely used in educational robotics projects.

One servo motor is assigned to each joint.

Total Servo Motors Required: 8

Although the servo motors were not modeled in Tinkercad, they were selected as the intended actuators for the mechanical design.

8. Initial Torque Calculation

The required joint torque can be estimated using the following equation:

Torque = Force × Distance

Assumptions

- Robot weight = **1 kg**
- Gravity = **9.81 N**
- Weight distributed equally among four legs

Force acting on one leg:

F = 9.81 / 4 = 2.45 N

Distance from the hip joint:

r = 0.05 m

Estimated torque:

T = 2.45 × 0.05 = 0.1225 N·m

Required torque:

≈ 0.13 N·m

This estimated torque is suitable for lightweight servo motors such as the SG90.

9. Stability and Center of Gravity

The center of gravity is positioned near the middle of the robot body to maintain balance during standing and walking.

The robot's stability is improved by:

- Symmetrical body design.
- Equal spacing between the four legs.
- Even weight distribution.
- A lightweight chassis.

These design choices reduce the risk of tipping over during movement.

10. Proposed Walking Method

The proposed walking method is the **Crawl Gait**.

Walking sequence:

1. Front Left Leg
2. Rear Right Leg
3. Front Right Leg
4. Rear Left Leg

Only one leg moves at a time while the remaining three legs remain in contact with the ground. This walking pattern provides excellent stability and is commonly used in simple quadruped robots.

11. Expected Mechanical Challenges

Several mechanical challenges may affect the robot's performance, including:

- Joint friction.
- Limited servo motor torque.
- Body vibration during walking.
- Foot slipping on smooth surfaces.
- Joint misalignment.
- Uneven weight distribution.
- Reduced stability on irregular terrain.

Future improvements can reduce these issues through better materials and more accurate mechanical design.

12. Skills and Experience Gained

This project was my first experience creating a three-dimensional (3D) mechanical model using **Tinkercad**.

During this project, I learned how to:

- Design a simple robotic dog using CAD software.
- Build a basic mechanical structure from simple geometric shapes.
- Design joints and leg mechanisms.
- Understand Degrees of Freedom (DOF).
- Select suitable servo motors.
- Estimate the required torque for robot joints.
- Analyze stability and center of gravity in a quadruped robot.